

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

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Purpose: A vertebra at the lumbosacral junction is named by counting. The gold standard counts caudally from C2 on whole-spine imaging¹; however, a lumbar case is planned on a lumbar study^{2,3}, T12 to S1, without C2. Abdominopelvic computed tomography (CT) holds that span plus the lowest ribs and pelvis, standing in for the preoperative view. Where a lumbosacral transitional vertebra (LSTV) alters the count, the local anatomy is ambiguous. Four rib-free vertebrae may be an L1 with a lumbar rib or an L5 assimilated to the sacrum, and six may be a sixth lumbar vertebra, a T12 with aplastic ribs, or a lumbarized S1⁴. CTSpinoPelvic1K asks whether local morphology resolves that ambiguity without the count. Two public label sets already covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients, never joined; this release joins them on one series and annotates the bones neither had. It provides 802 CT records, per-level ribs and femora, lumbar levels anchored on the lowest rib-bearing vertebra and S1, and classes for L6, T13, a separate S1 and lumbar ribs, so anomalies are recorded as themselves.

Acquisition and Validation Methods: Records pair CTSpine1K⁵ and CTPelvic1K⁶ labels on each patient’s bone-richest acquisition under a VerSe-native scheme. Validation covered geometric invariants (802/802 pass), rib–vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar cohort, planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles recumbent; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the
9 two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and
10 the lumbosacral transitional vertebra (LSTV) below, and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*. A thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly
14 wedged body, a reduced disc beneath it and hypoplastic facet joints, or conversely separated
15 from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc⁴.
16 They also change *how many* vertebrae each region contains, eleven or thirteen thoracic and
17 four or six lumbar.

18 Either change alone would be tractable. Together they make it difficult to state defini-
19 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
20 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
21 lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized away from the sacrum.
22 Each set yields the same count, and the morphology that would separate the readings is not
23 decisive to a human reader in every case. Whether it differs enough for a learned classifier
24 to separate them, or at least to assign each reading a probability, is the question this release
25 is built to support. Absent that, what remains is the count itself. The gold standard for
26 numeration in transitional anatomy is whole-spine imaging, counting caudally from C2¹,
27 which fixes how many vertebrae and how many rib pairs the column holds, and no spine-
28 limited acquisition can supply either count. LSTV is common, reported between 4% and
29 30% depending on definition, and wrong-level surgery is its most serious consequence.

30 Existing public collections cannot address this, and size is not the reason (Table I). Spine
31 datasets omit the pelvis; pelvic datasets do not number the vertebrae; TotalSegmentator,
32 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar
33 vertebra; and VerSe, the one collection with an L6 class, stops at the sacrum and carries
34 no ribs. No public collection can therefore record a six-lumbar spine together with both of
35 its borders, however good the segmentation, and a segmenter trained on them inherits the
36 gap, so that, faced with an enumeration anomaly, it must shift the whole column by a level

37 or absorb a vertebra into its neighbor. The size of that problem has now been measured on
38 CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and
39 an anomaly-aware extension of SPINEPS⁷ raised that to 99%⁸. Its weights label L6 and
40 T13 on CT, but its 1,536 training scans are in-house and unreleased, so the capability exists
41 as weights, not as data.

42 The intraoperative side has prior art of its own. LevelCheck registers the intraoperative
43 radiograph to the preoperative CT and projects the CT’s vertebral labels onto it^{9–11}, but
44 those labels are placed by hand and verified by the surgeon⁹, so the method assumes a
45 correctly labeled CT and does not supply one. Automating that step where the anatomy is
46 anomalous is what a dataset carrying the anomaly classes is for.

47 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
48 gives a class to each structure an enumeration anomaly produces, a sixth lumbar vertebra and
49 a rib borne by a lumbar vertebra. A scheme without those classes lacks a class vocabulary
50 expressive enough to describe anatomic anomalies.

51 The project began with a surgeon’s question, whether a vertebra can be named from
52 the anatomy in front of the reader, on the images a lumbar case is actually planned on.
53 By guideline those are lumbar studies^{2,3,12}, T12 to S1, and whole-spine coverage is reserved
54 for trauma with an identified injury¹³ and for deformity radiographs. The cohort suits
55 that question. An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it
56 holds the T12-to-S1 span of the planning study together with the lowest ribs and the whole
57 pelvis, and it lacks C2 exactly as that study does. Every record comes from one prospective
58 trial protocol, ACRIN 6664^{14,15}, which scanned 802 patients aged 50 and over, supine and
59 prone, on multidetector CT at 15 centers, on five scanner vendors and nine models, with
60 manufacturer, model and reconstruction kernel recorded per record. To our knowledge it
61 is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol; the
62 comparators pool several collections (CTSpine1K four, CTPelvic1K seven), are multi-site
63 by design (VerSe), or are routine clinical scans under many protocols (TotalSegmentator).
64 With the protocol held fixed, scanner effects on a model become measurable rather than
65 confounded. The price is an abdominal field of view, in which only the lowest thoracic levels
66 are present and C2 never is (Sec. V).

67 The gap was in plain sight. CTPelvic1K⁶ and CTSpine1K⁵, released months apart by
68 an overlapping author group, each annotated the COLONOG collection under radiologist

supervision, one the pelvis and the other the vertebrae, on the same patients, and no one joined them, although the join needs no new imaging and no new radiologist. It is not a file merge, because every patient was scanned prone and supine and neither source recorded which series it drew on, so each annotation had to be traced back to its own series (Sec. II.A). Done once, that crosswalk yields a spine-and-pelvis frame at 802 patients for the cost of the crosswalk, and the bones neither set had, ribs, femora, S1 and hardware, could then be annotated against it rather than from scratch. Two of the most-used public bone annotation sets were halves of one dataset; this release is the whole.

I.A. Vertebra labeling when the scan cannot settle the count

The limitation, stated plainly. No scan in this corpus contains C2, so the conventional count cannot be performed, which is a property of abdominal imaging, not of the annotation. A thirteenth thoracic vertebra and a lumbar rib are different phenotypes^{16,17}. The first is an additional rib-bearing segment, usually above five lumbar vertebrae, so the column holds 25 presacral vertebrae; the second is a rudimentary rib on a lumbar-type L1 in a column of the usual 24. Between the two borders used here, the last rib-bearing vertebra and the sacrum, a lumbar rib still leaves four rib-free vertebrae, as does a T13 in the rarer column of 24 with four lumbar vertebrae; a T13 above five lumbar vertebrae leaves five, and an aplastic twelfth rib leaves six. A *stump rib* (a hypoplastic twelfth rib, a cardinal indicator of a thoracolumbar transitional vertebra) is still a rib, so it leaves the count unchanged, and only its length records the anomaly. Rib anomalies travel with the lumbosacral border; on whole-spine CT, hypoplastic twelfth ribs occur with sacralization and lumbar ribs with lumbarization¹⁸. Without C2, a count of four therefore cannot separate a lumbar rib from a cranially shifted T13 or from an L5 assimilated to the sacrum, and a count of six cannot separate an aplastic twelfth rib from a sixth lumbar vertebra.

What the release does in addition. Every vertebra carries the identifier its radiologist-sourced annotation assigned, corrected where the source was wrong; those labels are the ground truth. Alongside them, every released measurement is also expressed against the two landmarks present in essentially every abdominal scan, the sacrum below and the lowest rib-bearing vertebra above, so that the measurement is positional rather than nominal, a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it²⁴.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ⁵	1,005									
CTPelvic1K ⁶	1,184									
VerSe ²⁵	374									
RibSeg v2 ²⁶	660									
TotalSegmentator ²⁷	1,204									
CTSpinoPelvic1K	802									

length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes with the name a reader gives the junction, so cases that disagree about the name remain comparable.

But the vertebrae themselves are known to differ. Counting is not the only route to an identity. Level-specific morphometry is long established. Pedicle width and height change systematically from the thoracic to the lumbar spine,¹⁹ vertebral body, endplate and canal dimensions differ by level,^{20–22} and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.^{4,23} If those differences hold at the boundary itself, the phenotype is decidable locally, and on this corpus it does hold. Separating T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is removed because raw millimeters separate thoracic from lumbar trivially and teach nothing about the junction, where the two are nearly the same size.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both sources draw part of their imaging from the TCIA CT colonography collection^{14,15}. CTSpine1K’s 1,005 volumes come from four collections and CTPelvic1K’s 1,184 from seven; COLONOG is the only collection common to both and the only one acquired under a single protocol, so this release is its COLONOG subset, 782 patients with a CTSpine1K vertebral

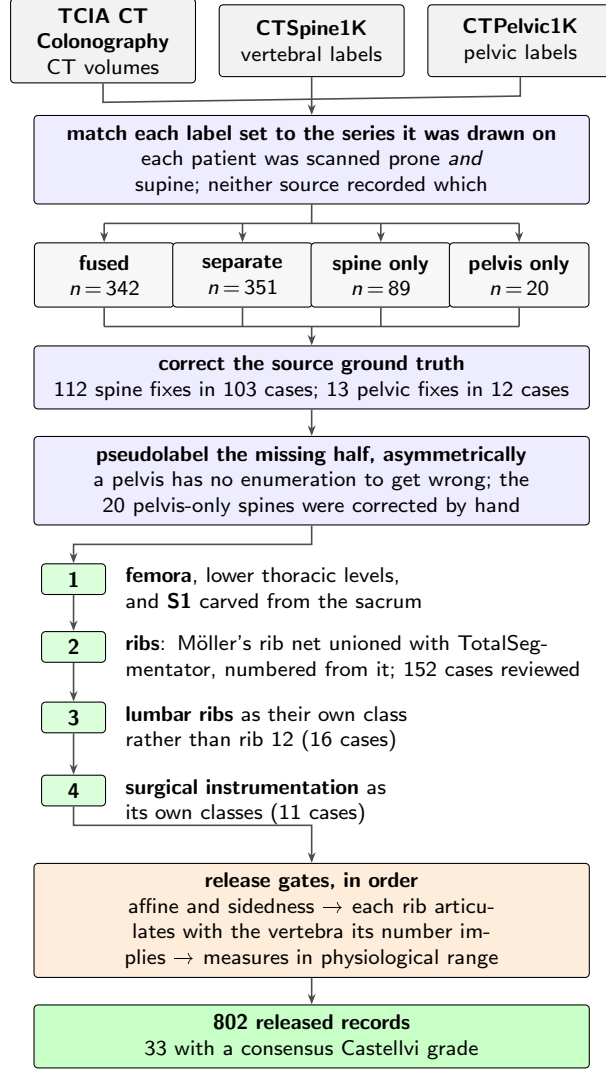


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

118 annotation and 20 with a CTPelvic1K pelvic annotation only, 802 in all. Neither source
119 released the mapping from an annotation to the specific CT series it was drawn on.

120 That mapping matters because every patient in CT COLONOGRAPHY was scanned
121 *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer.
122 Turning a patient from supine to prone alters lumbar lordosis and the alignment of each
123 segment against the next, so outlines drawn on one series are the wrong *shape* for the other
124 and no rigid realignment recovers them. The crosswalk (Fig. 1) therefore resolves each
125 annotation to a patient and then, within that patient's volumes, to the series whose bone
126 *agrees* with it, where candidates are thresholded at bone attenuation and scored by overlap
127 with the mask.

II.B. Fused and split records

CTSpine1K and CTPelvic1K each annotated a patient independently, so their labels are not guaranteed to come from the same CT series (Fig. 1). This turned out to matter for 351 patients, nearly half the cohort, because the two collections had labeled different acquisitions of the same person, one prone and one supine, with neither recording which. Because posture changes spinal alignment, a spine label from one acquisition cannot simply be paired with a pelvic label from the other. These 351 records are released but held out of any analysis that assumes a single posture; two labeled postures of one patient are a resource, not a defect.

II.C. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the safe direction, since the sacrum and innominate bones carry no enumeration to get wrong, and quality is reported as held-out performance on the *pelvic_only* records, withheld from training for that purpose. The reverse direction carries the whole problem this dataset addresses, since a pseudolabeled spine must commit to a count and on a transitional vertebra commits to the commoner reading; those 20 records were corrected by hand, tractable at that scale.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller’s binary rib network²⁸ reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. On this corpus the segmentation itself failed on ribs that were *partially outside* the field of view. A rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. In a colonography cohort the lower ribs are routinely clipped, and those are the ribs the count depends on.

TotalSegmentator²⁷ does segment clipped ribs and supplies the per-rib numbering that a

156 binary network cannot. Its characteristic failure is at the other end of the bone, where the
157 most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral
158 joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it
159 articulates with, so the rib-vertebra incidence check (the quality-control gate on the rib class
160 labels, which matches each rib to the vertebra its number implies) has nothing to test.

161 The two were therefore unioned (Fig. 1), TotalSegmentator supplying the numbering and
162 the medial-clipped cases, Möller the detection where its output is more complete.

163 The result is a pseudolabel whose review was triaged rather than exhaustive. A label-
164 based rule flagged every record in which one rib bone carried more than one identifier, or in
165 which a rib failed to reach the vertebra it articulates with, since a single bone carrying two
166 numbers is a numbering error by construction, whatever the anatomy. The rule selected 152
167 records; 148 were reviewed and corrected, and four were not and are identified in the release.
168 Reviewers were medical students working through a student research consortium²⁹, trained
169 against a written labeling protocol, with every correction recorded against the annotator
170 who made it. Records passing the rule were not individually inspected, so the layer only
171 guarantees freedom from the failure modes the rule detects, no more. The residual rib-
172 vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

173 II.E. Label scheme and counting anchors

174 The scheme is VerSe-native. Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12
175 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure
176 takes a fixed identifier above that range, S1 carved from the sacrum (29), hips (30–31),
177 femora (32–33), thirteen ribs per side (34–46 left, 47–59 right), lumbar ribs (60–61) and
178 surgical hardware (62–68). The space is contiguous. Rib 13 is the true rib of a T13 and is
179 empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the
180 distinct phenotypes they are.

181 Two anchors give every record the same positional frame, and Fig. 2 shows both, the
182 lowest rib-bearing vertebra above, **S1** carved from the sacrum below. S1 supplies the sacral
183 endplate landmark that sacral slope and pelvic incidence are measured from, and with both
184 brackets present L5 and L6 are distinguishable where a spine-limited view alone would not
185 be.

186 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never altered;
187 S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the
188 sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis measured directly
189 rather than assumed, so as to remove the patient’s rotation within the scanner. That
190 rotation is not negligible, with a median of 4.5°, a maximum of 16.1°, and 508 of 801 records
191 above 3°. The cut preserves the sub-division volume of the preceding release, changing
192 the orientation of the S1/S2 boundary and not how much sacrum is called S1; per-record
193 geometry ships with the archive.

194 VerSe²⁵ carries L6, whose identifier this release keeps, so its L6 *is* VerSe’s.

195 **A rib on a lumbar body receives its own class**, and this is the one class here with
196 no published counterpart. A lumbar rib and a stump rib are the same object under two
197 counts, so assigning it a number would commit the label to one reading of an enumeration
198 anomaly. A scheme that numbers every rib 1–12 has nowhere to put a thirteenth, so the
199 annotator must either call it rib 12, asserting the vertebra beneath it is thoracic, the very
200 question at issue, or discard it. TotalSegmentator carries ribs 1–12 per side and no such
201 class, and VerSe’s T13 is a vertebra rather than a rib. This scheme can record both, VerSe’s
202 T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for
203 a lumbar-type body bearing a rudimentary rib, so either reading is recorded as itself. No
204 record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies.
205 Where a border vertebra could not be settled by its readers the label stands as read, and
206 reclassifying those records with a classifier trained on local vertebral morphology (Sec. V)
207 is the intended path for a later release.

208 Figure 2 shows why the interval is the primitive and the label is derived. Among the
209 records with four rib-free vertebrae, some reach that count from above, through a rib on
210 L1, and some from below, through a lumbosacral segment fused to the sacrum, in Fig. 2(c)
211 together with stump twelfth ribs; the count alone cannot tell them apart, and the release
212 can because ribs, rib lengths and the S1 boundary are all labeled.

213 Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right,
214 too few to speak to side preference.

215 Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the
216 eighteenth was not graded. The source transitional label (pelvic where CTPelvic1K flagged
217 one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and

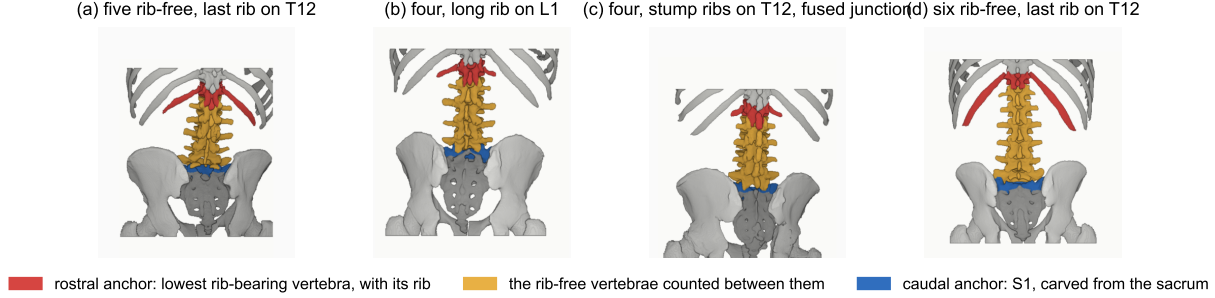


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four. In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb) and carries the S1 identifier.

semi-sacralization in one, and is unremarkable in the ungraded record. A spine can carry six lumbar-type vertebrae and still be read as a sacralization, depending on where the reader started the count.

Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`, `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by counting identifiers 25 and 60–61.

II.F. Computational tools

Access. Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters. Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator²⁷ (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s²⁸ released weights, applied through the nnU-Net v2 framework³⁰. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold, so that each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (the repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version). Image handling throughout used NiBabel and SciPy; no

image is resampled or reoriented when a label is written, so every label shares its image's grid and affine exactly.

Generative artificial intelligence. A large language model (Claude, Anthropic) assisted with drafting and editing the manuscript text and with writing the analysis and figure code, all under author review. It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.

II.G. Validation

Validation has three layers. Geometry is checked first, because a measurement taken on a corpus with a geometric fault still returns a plausible-looking number.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left-right axis read from the affine rather than assumed.

The check compares *each sided pair separately*. Testing the ribs alone passed all 802 records while four carried `left.hip` on the patient's right, and pooling all sided structures into one test still missed one of those four, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib-vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined. Both offsets are field-of-view truncations on individually characterized cases, one at +1 level and one at -1, reported rather than suppressed.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting

TABLE II Derived measures against a standing reference ($n = 260$)³¹; this cohort is supine.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

error of exactly the kind this dataset exists to expose.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*³¹ (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

II.H. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade³² in the released manifest, 33 cases typed I–IV with the *a/b* unilateral–bilateral qualifier. The grade and the count are *different axes*. Grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)⁴, is the one that matters clinically, and four of the six disagreements were III against II.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. A cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does.

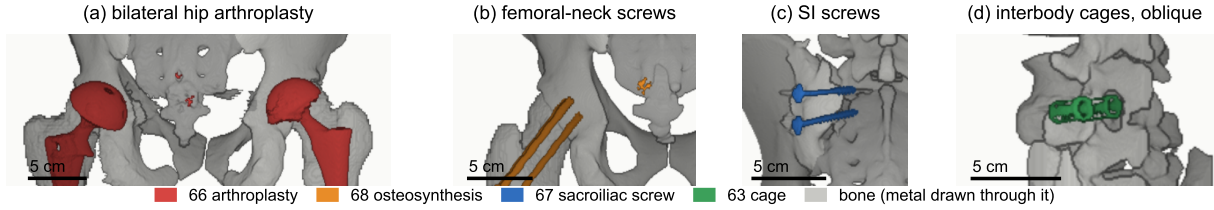


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67, $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

286 *II.I.0.a. Detection over-calls by a factor of five.* A 2500 HU threshold inside a shell
 287 around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident
 288 physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and
 289 reconstruction artefact. Attenuation does not separate the two groups (both saturate);
 290 site and volume do, every confirmed implant at $2,586 \text{ mm}^3$ or above and every rejection at
 291 $1,768 \text{ mm}^3$ or below.

292 *II.I.0.b. The subtype block had to be extended.* Identifiers 62–65 name spinal instrumen-
 293 tation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so
 294 **66 arthroplasty**, **67 sacroiliac screw** and **68 osteosynthesis** were added. Osteosynthe-
 295 sis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule
 296 calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis,
 297 one sacroiliac fixation and one pair of interbody cages (Fig. 3).

298 *II.I.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the
 299 implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than
 300 adding them. In eight of these cases the femoral head that pelvic incidence and tilt are
 301 measured from is an implant.

302 III. DATA FORMAT AND USAGE NOTES

303 Each record is a gzipped NifTI image/label pair sharing an identical 4×4 affine, canon-
 304 icalized to PIR, requiring no resampling. Masks are NifTI rather than DICOM, which is
 305 what volumetric segmentation tooling expects.

306 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag, so every validation fold carries three sacralization and three lumbarization records. With 15 of each across 802 that is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

Table III lists the key data types and metadata against the fraction of records for which each is populated.

The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

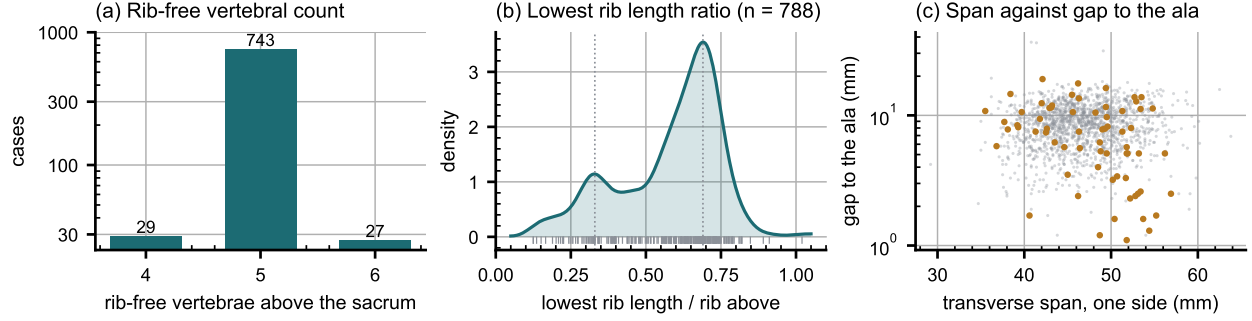


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM's *other* value and 53 no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven outside the female/male dichotomy are excluded from sex-stratified measures.

Count-free measures (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of the rib above, marks a stump twelfth rib in 98 of the 788 records with a measurable pair (12.4%), against 12.6% on whole-spine CT¹⁸; 16 records carry a lumbar rib (2.0%, against 1.5%). The co-occurrence reported there, stump ribs with sacralization and lumbar ribs with lumbarization, is testable here only where the junction was read; stump ribs accompany four of the 14 source-labeled sacralizations and 94 of the other 758 records (odds ratio 2.8, $p = 0.09$, Fisher's exact test), and no lumbarization carries a lumbar rib.

Level gradients and spinopelvic measures (Fig. 5). Vertebral dimensions increase caudally; pelvic incidence does not vary with age while pelvic tilt does, the signature of a pelvis retroverting as a spine ages.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.³³ L1 trabecular attenuation is reported at the published standard site.

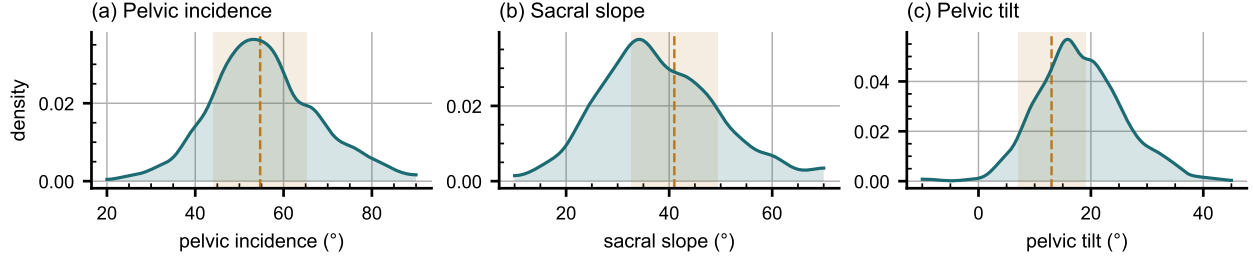


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)³¹.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable against the lumbar column rather than in isolation, among them the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it, since no trajectory can be planned across a junction whose segments cannot be named.

A proxy for the preoperative view. No public preoperative lumbar CT cohort exists that we know of, and VerSe lacks the sacrum and pelvis (Table I), so this release is the closest available stand-in for the view a lumbar case is planned on. It holds the T12-to-S1 span with the lowest ribs and the pelvis, 377 of its 802 records were acquired prone, the position of posterior lumbar surgery, and it carries classes for the anomalies that make level identification fail. It is a screening population at colonography dose with standard soft-tissue kernels, not a surgical series, and it supports method development for level identification and instrumentation geometry, not validation of a planning decision.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this cohort was assembled around.³⁴ Per-level body height, canal width, endplate width, transverse process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained

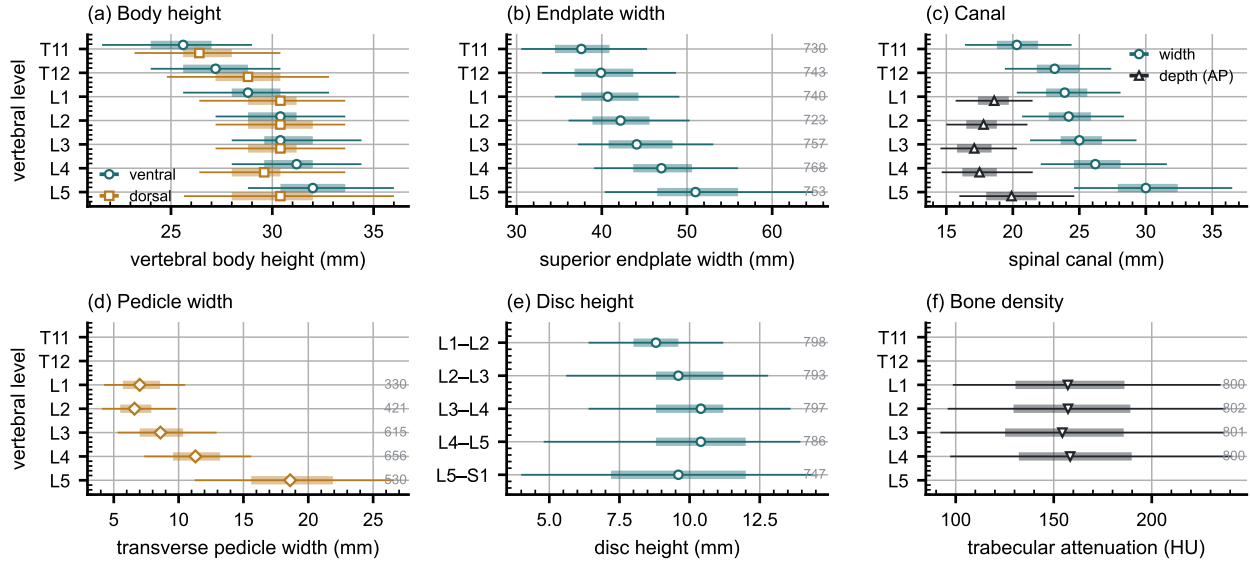


FIG. 6 Morphometry by level, as median, interquartile range and 5th–95th percentile, with n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

on them is not learning the convention that fails.

Reference morphometry with its spread. The values a surgeon consults for vertebral body height, canal dimensions and pedicle width come from cadaveric series of a few dozen specimens, plotted as single values with no indication of spread.²² They cannot tell a reader whether the patient in front of them is unusual, because they never show what usual looks like. The same measures are given here from 802 records with the distribution attached (Fig. 6), reproducing two properties of the classical figures (body height crosses over, dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply into the lower lumbar spine) and adding the width of each distribution, which is the part needed to judge an individual.

Patient-specific models. Planning is moving toward patient-specific biomechanical models rather than population norms.³⁵ Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction; in 351 patients two postures give a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat.

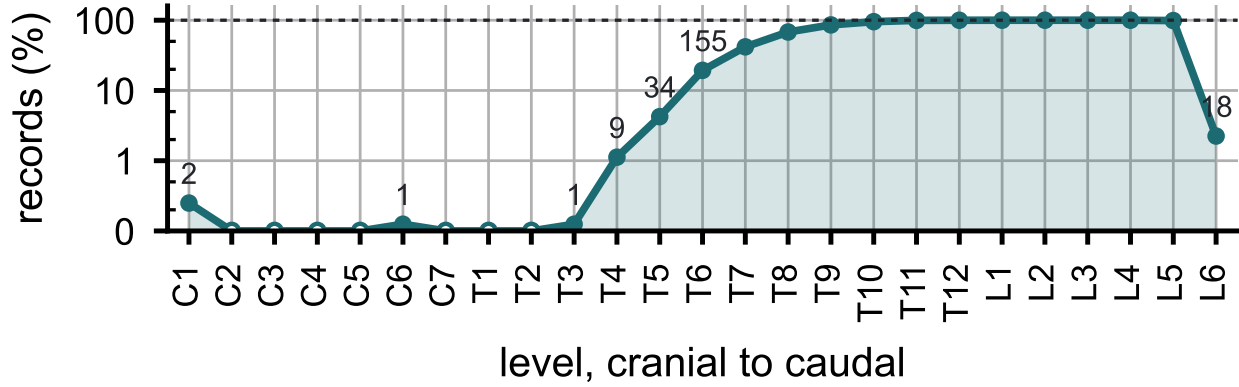


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

379 The cohort is a colorectal screening population aged 50 and over, so its distributions should
 380 not be read as representative of a surgical one.

381 *Strength of the labels varies by structure, and the release does not average over that.*
 382 Vertebral labels derive from radiologist-supervised source annotations, corrected where those
 383 sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib
 384 layer is a pseudolabel whose human review was triaged by an automated rule rather than
 385 exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional
 386 labels derive from two independent sources and are not uniformly adjudicated, which is
 387 precisely why the measures reported here are count-free, and the Castellvi grades are a
 388 consensus of two radiology resident physicians. S1 is carved from the sacrum by an automatic
 389 estimate of the S1–S2 boundary, and in 100 records that estimate is implausible, making
 390 S1 more than half the sacrum’s height or less than fifteen percent of it. Those records are
 391 flagged in the quality-control table; exclude them from any measurement that depends on
 392 the sacral endplate.

393 *Structures are not universally present.* Sacrum, hips and femora are present in all 802
 394 records; one record carries no S1 and one no T12, each outside the field of view. Nine records
 395 carry no L5 identifier because their source annotation counted four lumbar vertebrae, all nine
 396 are Castellvi IIIb, and the fused transitional segment carries the S1 identifier as the caudal
 397 anchor; the manifest names them.

VI. DISCUSSION

Comparison with related material. Against the collections in Table I, the contribution is the crosswalk placing spine and pelvis on the same series, together with classes for the anatomy an enumeration anomaly actually produces. VerSe^{24,25} comes closest in intent, enriching for transitional anatomy and grading by Castellvi, but it does not segment the sacrum, so the structure a grade describes has no mask; this release segments it, as L6 or as a separately carved S1. Extending VerSe with a sacrum would not have supplied the pelvis, femora or prone acquisitions, and here both label sets already existed. What this release adds over TotalSegmentator²⁷ is not more anatomy per scan but the classes an anomaly needs to be recorded accurately.

Limitations. The cohort comes from one acquisition protocol of a colorectal-screening population aged 50 and over, so how well it generalizes to a pathologic population is untested. Human review of the rib layer was triaged by an automated rule, so it guarantees only the absence of the failures that rule detects.

Future directions. Imaging beyond this protocol, and the C2 anchor it cannot supply, are the natural next additions; applying the released weights to VerSe would add the S1 anchor and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole cohort, rather than of the 33 flagged records, would test the co-occurrence of rib and lumbar anomalies at full power. The S1 carve should be replaced by one that follows the anatomy, so that the caudal anchor lands consistently across normal and sacralized junctions, and whether the fused segment in the nine four-lumbar records is a lumbar-type body is a question of pelvic morphology, for deep-learning morphometrics of the pelvis read in isolation.

VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own rather than recording it as something it is not. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2, which is every record here.

427 DATA AVAILABILITY

428 The release described here is v10, permanently archived at [https://doi.org/10.5281/](https://doi.org/10.5281/zenodo.22642578)
429 [zenodo.22642578](https://doi.org/10.5281/zenodo.22642578); the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves
430 to the current version. The archive carries the loader, the label scheme, the quality-control
431 tables and the analysis code under an open-source licence. The repository URL identifies
432 the authors; it is on the title page and will be restored here for the camera-ready version.

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435 view policy. This work received no funding.

436 CONFLICT OF INTEREST

437 The authors have no relevant conflicts of interest to disclose.

438 ETHICS

439 This work uses publicly available, de-identified imaging and annotations derived from it.
440 The authors' institutional review board determined that it is not human participant research
441 under 45 CFR 46 and requires no oversight.

442 REFERENCES

- 443 ¹J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and
444 associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-
445 018-5554-8.
- 446 ²Expert Panel on Neurological Imaging, T. A. Hutchins, M. Peckham, et al., ACR Ap-
447 propriateness Criteria low back pain: 2021 update, *J. Am. Coll. Radiol.* **18**, S361–S379
448 (2021), doi:10.1016/j.jacr.2021.08.002.

- ³American College of Radiology, ACR–ASNR–SABI–SSR practice parameter for the performance of magnetic resonance imaging (MRI) of the adult spine, revised 2023 (Resolution 6), American College of Radiology, Reston, VA, 2023.
- ⁴G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.
- ⁵Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.
- ⁶P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- ⁷H. Möller, R. Graf, J. Schmitt, et al., SPINEPS: automatic whole spine segmentation of T2-weighted MR images using a two-phase approach to multi-class semantic and instance segmentation, *Eur. Radiol.* **35**, 1178–1189 (2025), doi:10.1007/s00330-024-11155-y.
- ⁸H. Möller, H. Schoen, R. Graf, et al., VERIDAH: solving enumeration anomaly aware vertebra labeling across imaging sequences, 2026, arXiv:2601.14066. doi:10.48550/arXiv.2601.14066.
- ⁹Y. Otake, S. Schafer, J. W. Stayman, et al., Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery, *Phys. Med. Biol.* **57**, 5485–5508 (2012), doi:10.1088/0031-9155/57/17/5485.
- ¹⁰S.-F. L. Lo, Y. Otake, V. Puvanesarajah, et al., Automatic localization of target vertebrae in spine surgery: clinical evaluation of the LevelCheck registration algorithm, *Spine (Phila. Pa. 1976)* **40**, E476–E483 (2015), doi:10.1097/BRS.0000000000000814.
- ¹¹T. De Silva, S.-F. L. Lo, N. Aygun, et al., Utility of the LevelCheck algorithm for decision support in vertebral localization, *Spine (Phila. Pa. 1976)* **41**, E1249–E1256 (2016), doi:10.1097/BRS.0000000000001589.
- ¹²D. S. Kreiner, W. O. Shaffer, J. L. Baisden, et al., An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update), *Spine J.* **13**, 734–743 (2013), doi:10.1016/j.spinee.2012.11.059.
- ¹³American College of Surgeons Committee on Trauma, ACS TQIP best practices guidelines in imaging, American College of Surgeons, Chicago, 2018.

¹⁴K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

¹⁵K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.

¹⁶A. M. Du Plessis, L. M. Greyling, and B. J. Page, Differentiation and classification of thoracolumbar transitional vertebrae, *J. Anat.* **232**, 850–856 (2018), doi:10.1111/joa.12781.

¹⁷A. M. Poolman, Q. Wessels, A. Van Schoor, and N. Keough, Thoracolumbar transitional vertebrae: quantitative differentiation and associated numeric variation in the vertebral column using skeletal remains, *J. Anat.* **243**, 311–318 (2023), doi:10.1111/joa.13865.

¹⁸S. Nagata, C. Yamada, M. Sawano, et al., Lumbosacral transitional vertebra on whole-spine CT: prevalence and association with rib abnormalities, *Skeletal Radiol.* **54**, 2169–2177 (2025), doi:10.1007/s00256-025-04952-z.

¹⁹M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and lumbar pedicles, *Spine (Phila. Pa. 1976)* **12**, 160–166 (1987), doi:10.1097/00007632-198703000-00012.

²⁰M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.

²¹M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy, *Spine (Phila. Pa. 1976)* **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.

²²E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

²³R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on MRI: role of the iliolumbar ligaments, *AJR Am. J. Roentgenol.* **187**, W59–W65 (2006), doi:10.2214/AJR.05.0415.

²⁴H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.

²⁵A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.

²⁶L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024),

doi:10.1109/TMI.2023.3313627.

²⁷J. Wasserthal et al., 'TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.

²⁸H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

²⁹[Reference to the authors' own work, removed for double-anonymized review.]

³⁰F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

³¹R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

³²A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.

³³P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.

³⁴N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, *Surg. Neurol. Int.* **12**, 286 (2021), doi:10.25259/SNI_402_2021.

³⁵E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, *J. Clin. Med.* **15**, 2584 (2026), doi:10.3390/jcm15072584.

FIGURE CAPTIONS

Figure 1. How the dataset was built. n counts scans, or corrections, at each step.

Figure 2. The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four. In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb) and carries the S1 identifier.

Figure 3. Instrumentation in the release, one exemplar each, drawn through bone. (a)

543 Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67,
544 $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

545 **Figure 4.** Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b)
546 Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to
547 the ala; transitional labels marked.

548 **Figure 5.** Spinopelvic measures against standing reference bands (mean $\pm$ SD)³¹.

549 **Figure 6.** Morphometry by level, as median, interquartile range and 5th–95th percentile,
550 with n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e)
551 Disc height. (f) Trabecular attenuation.

552 **Figure 7.** Records carrying each vertebral level, cranial to caudal. Counts per identifier in
553 Table S1.